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ENST200  
Section 0102

## Personal Soil Lab Report

### Introduction

Performing an extensive soil analysis on one soil requires being able to quantify the physical, chemical, and biological properties. Understanding how the soil properties work together and evaluating the correlation between the data explains how to achieve the best soil based on various land management strategies. Achieving reliable data can show how factors like Cation Exchange Capacity determines nutrient retention, or how Organic matter and Active Carbon speak about the soil health and carbon potential which can help you identify how to adjust these factors to have a healthier soil. During the past four months, my classmates and I conducted research on personal soil samples that we collected so that we could analyze the data, make predictions and management plans based on how our data compared to the rest of the class. There were three main comparisons that were made between the data: percent clay versus cation exchange capacity (CEC), active carbon versus aggregate stability, and soil organic matter versus volumetric water content. I am hypothesizing that the relationship between percent clay and CEC will be a positive linear relationship because clay particles possess high specific surface area and negative charges which provide an interface for plant cations. I am also hypothesizing that active carbon and agg. stability will have a positive linear relationship because microbes easily metabolize carbon and microbes produce glue like substances that bind particles. My last hypothesis remains the same that I predict to see a positive linear relationship because organic matter has high porosity which helps retain water.

## Site Description

My personal soil was taken from 8430 Baltimore Avenue which is the apartment building Tempo, College Park. The specific site is about 40 feet from Paint Branch Creek and 100 feet from Tempo apartment buildings. The site is currently a frequently flooded riparian zone labeled a Codorus or Hatboro series. The current vegetation is some deciduous tree cover and a mixture of different grass types. In the past, this area has functioned as a floodplain which led to the deposition of alluvial sediments as time passed.



Figure 1. Pictured is the site of the personal soil sample and specific area is circled in red.

## Methods

The personal soil sample was taken on January 12th, 2026 at a depth of 0-6 inches or 0-15 cm. My soil was classified using the Web Soil Survey as Fluvaquentic Dystrudepts / Endoaquepts. Texture by feel was performed using a water and soil mixture to follow the texture by feel flow chart on page 27 of the laboratory manual. We then used that knowledge to estimate the percent of sand, silt, and clay by using the textural triangle on page 30 of the laboratory manual based on the result of the flow chart. Texture by hydrometer was performed by agitating the soil in water and taking various hydrometer readings of the suspended soil to show how many aggregates of each size were still suspended. Using these readings and calculations on

page 74 of the laboratory manual we calculated percent sand, silt, and clay. Color was determined by moistening the soil and comparing the soil to a Munsell color book that was given to the class to measure hue, value, and chroma. Aggregate stability was measured by placing a larger aggregate of soil in a small pool of water and using a net to submerge the soil. We then timed the lab and took measurement based on stability ratings given to us on page 86 of the laboratory manual. Water content was measured by comparing oven dry soil to air dry soil by using a microwave oven. Organic matter was measured by heating soil in a crucible to destroy the organic matter and taking before and after weights. pH of soil in water and KCl were measured by adding water and KCl to each and NaOH to both to our soils and taking a pH reading with a pH electrode. CEC was measured using gentian violet and a battery to show the exchange of cations between the soil and taking the absorbance to achieve CEC rating. Active Carbon was measured by adding  $\text{KMnO}_4$  to soil and shaking, then adding to a cuvette and measuring the absorbance to use the equation on page 178 of the laboratory manual. Plant-available phosphorus was measured using charcoal and Mehlich 1, to create a filtrate that we adjusted the pH of and read an absorbance rating.

## Results

### Personal Soil Data

Textural Class (by feel, by hydrometer): Sandy Clay Loam, Silty Clay Loam

% Sand (by feel, by hydrometer): 60%, .49%

% Silt (by hydrometer): 32%

% Clay (by feel, by hydrometer): 30%, 19%

Color (moist): 10YR, 4, 6

Agg. Stability rating (1-6): 1.75

Water content at -5kPa: 0.2  $\text{cm}^3/\text{cm}^3$

Total organic matter loss on ignition: 5.3%

Active Carbon: 878.4 mg/kg

pH (water, KCL): 6.48, 5.61

Cation Exchange Capacity: 11.9 cmolc/kg

Plant-Available P: 5 mg/L

Soil survey map unit designation: Codorus and Hatboro soils, frequently flooded

Soil taxonomy classification: Fluvaquentic Dystrudepts, Fluvaquentic Endoaquepts, and Aquic Hapludults

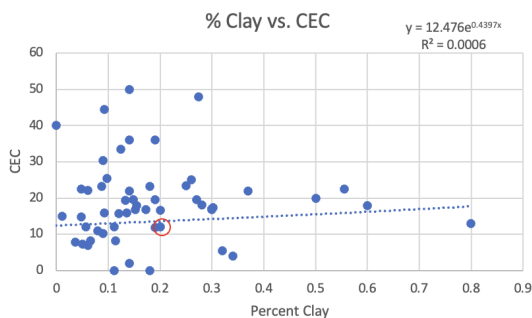


Figure 2. The relationship between percent clay in soil (%) and cation exchange capacity (cmolc/kg). The trendline is exponential relations to the graph data. The slope of the line is  $y = 12.476e^{0.4397x}$  and the  $R^2$  value is 0.0006.

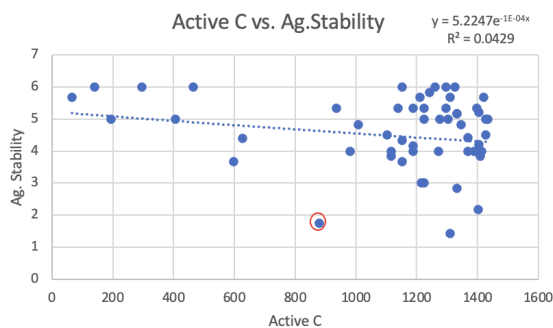


Figure 3. The relationship between active carbon in soil (mg/kg) and aggregate stability (1-6). The trendline is exponential relations to the graph data. The slope of the line is  $y = 5.33476e^{-1E-04x}$  and the  $R^2$  value is 0.0429.

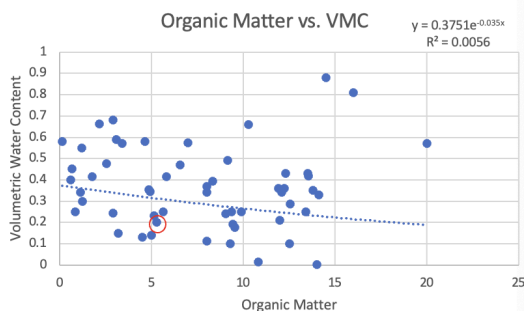


Figure 4. The relationship between organic matter in soil (%) and volumetric water content (cm<sup>3</sup>/cm<sup>3</sup>). The trendline is exponential relations to the graph data. The slope of the line is  $y = 0.3751e^{0.035x}$  and the R<sup>2</sup> value is 0.0056.

## Discussion

While the chemical properties of my soil suggest that the soil has a high potential for nutrient retention and good biological activity, the physical properties are not showing that they would be suitable for many land uses. My soil organic matter is 5.38% which is low out of the class, this is likely due to the floodplain area and anaerobic conditions from that. This frequent flooding disrupts the stability of the soil where mine showed to have one of the lowest overall ratings of a 1.75. The pH of 6.48 is optimal for nutrient availability but my phosphorus is low at 5 mg/L and active carbon is moderate compared to the class 878.4 mg/kg. My CEC vs. clay content graph is showing that my soil is below the trendline which indicates that the clay that my soil does have is not holding onto those actions that it should. It may be because of flooding and washing away those ions that would be sticking to the clay. My aggregate stability is way below the trendline as well which may indicate that the aggregates are falling apart faster than microbe byproducts from the active carbon can put them back together. My organic matter in my soil is acting as the sponge for the water content instead of the clay. Overall my soil is contradicting itself in many ways and most of these effects are most likely coming from the frequent flooding potential of the soil.

My soil could act as many things when it comes to land management. This soil would thrive as a natural wetland. The “Endoaquept” classification and high VWC make it perfect for wetland conditions. It is a natural environment for many nutrients and carbon for wetland plants. A moderate spot where this might or might not fit would be used in agriculture as pasture. While nutrients are decent, the low aggregate stability and high silt content make it prone to puddling.

The heavy livestock in the area will compact the soil and leave it as a mess even though it has great nutrients for grasses for these livestock. The major weakness that this soil has is any type of infrastructure of building. The “Fluvaquentic” flooding risk of this soil and high organic matter shows that it is highly compressible. This soil is the worst suited for foundations and any type of septic drainage due to the high water table.

While analysing the  $R^2$  values of the class data the strongest trend and  $R^2$  value is supported by the data of percent clay versus CEC. The second graph  $R^2$  is rejected because of the clear low  $R^2$  value and variation in data. The last graph  $R^2$  is supported because it shows a moderate to strong correlation between the data. There were probably many sampling handling errors within our data that did not make it reliable. The nature of the classroom makes it hard to have perfect data and the mistakes were still recorded which skews data and trendlines.

## References

R.R. Weil. 2021. Laboratory Manual For Introduction Soils. 10th ed. Kendall Hunt Publishing Company.

Soil Survey Staff, Natural Resources Conservation Service, United States Department of Agriculture. (2026). Web Soil Survey. <https://websoilsurvey.nrcs.usda.gov/>

## **Honor Code Pledge**

"On my honor, I have neither given nor received aid on this assignment."

Signature: Anna Swann